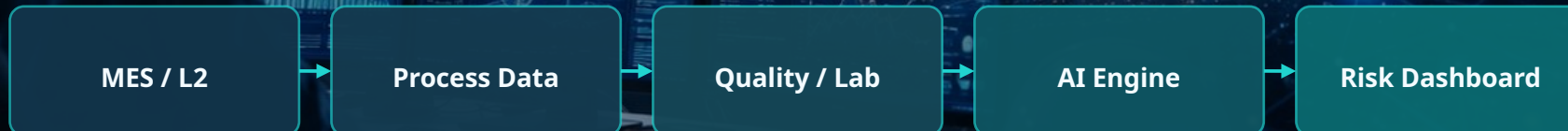


PlantProcess IQ

AI-based process-to-quality intelligence for manufacturing plants

Connect production, process, inspection and quality data. Discover defect correlations. Predict risk earlier. Explain why.



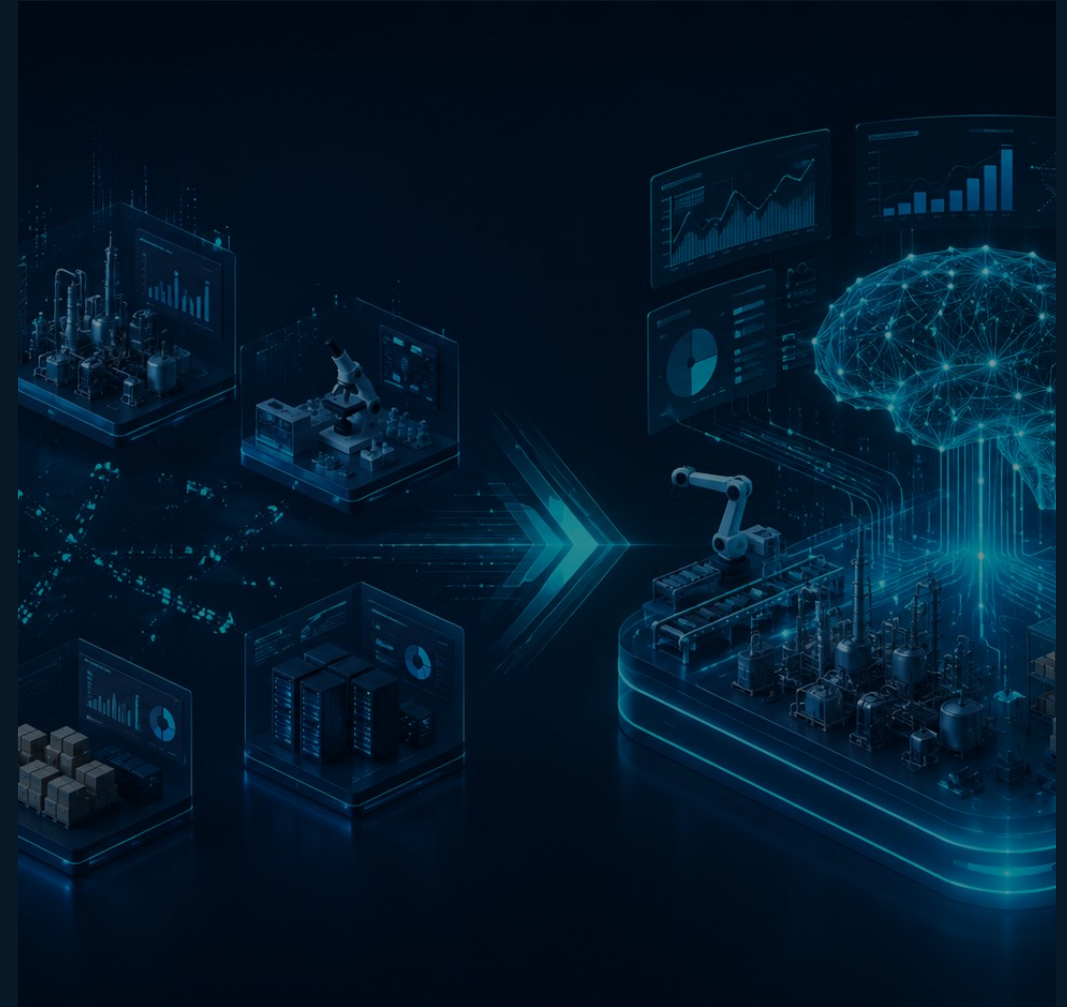
Generic - Read-only - Pilot-ready

The manufacturing data problem

Most plants have the data. They do not have connected quality intelligence.

- Production, process, lab, inspection and downtime data live in separate systems.
- Quality investigation still depends on Excel, screenshots, SQL and tribal knowledge.
- Traditional dashboards show what happened - not which conditions likely caused it.
- Late detection creates scrap, rework, downgrades, claims and expensive delays.

The opportunity: turn fragmented plant data into a repeatable quality decision engine.



From data silos to quality intelligence

PlantProcess IQ is an intelligence layer above existing MES, L2, SCADA, historian, lab and quality systems. It does not replace them - it reads, connects, learns and explains.

Read existing data

Build material / batch
genealogy

Detect data-quality gaps

Find process-to-defect
signals

Predict risk + explain

Vision

Act before defects become
expensive.

Positioning

AI co-pilot, not black-box autopilot.

What the application will deliver

A practical feature map from safe data ingestion to risk-based decision support.

Connect

CSV/Excel, SQL exports, APIs, approved read-only source access

Detect

Missing IDs, broken genealogy, duplicate records, timestamp gaps

Normalize

Generic canonical model: material, equipment, process, parameters, quality

Analyze

Defect rate by parameter range, equipment, route, grade, shift and crew

Trace

Material/batch genealogy across splits, transformations and inspections

Predict

Risk score, risk class and suspected contributors with explainability

Start safe with historical exports and read-only access. Grow toward scheduled extraction, dashboards and prediction.

What customers can expect

PlantProcess IQ turns quality investigation from manual detective work into a repeatable, evidence-based process.

Faster

root-cause investigation

Earlier

quality-risk warning

Better

process-window learning

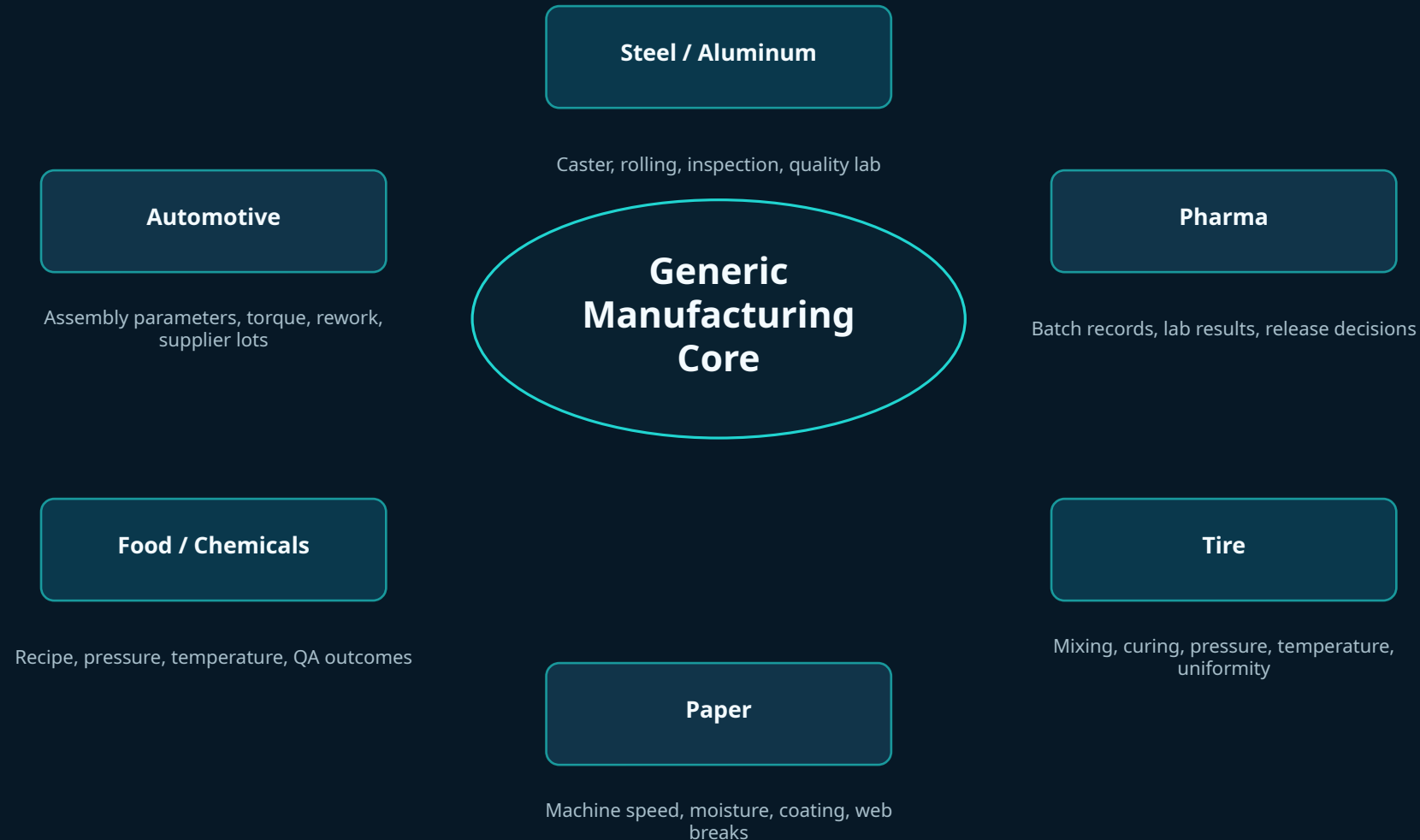
Safer

read-only pilot path

- Reduce scrap, rework, downgrade, claims and delayed release.
- Expose hidden data-quality issues before analytics becomes unreliable.
- Rank suspected contributors - equipment, route, parameter window, grade, shift or crew.
- Give process engineers a co-pilot with evidence, not unsupported certainty.

Built for plants with complex process-to-quality relationships

One generic core; multiple industry configuration packs.



Best first buyers: quality leaders, plant managers, manufacturing IT/OT, digitalization teams and system integrators.

Pilot story: from EAF/LF to Caster to HSM quality risk

Example hidden pattern

High casting speed fluctuation + low superheat + risky mould/powder combination can raise surface-crack risk downstream.

Business result

Flag high-risk material before expensive downstream processing.

EAF / LF
chemistry + treatment

Caster
speed + superheat + mould

HSM
reheat + rolling force

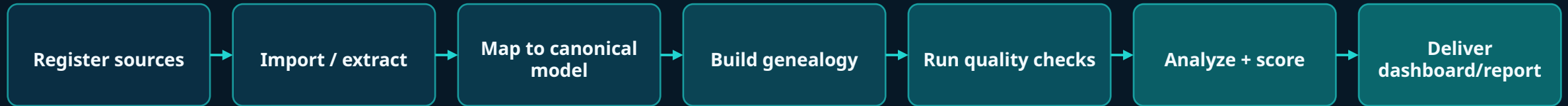
Inspection / Quality
defects + decisions

AI Co-pilot
risk + suspected drivers



A business-friendly process flow

Safe first. Auditable. Designed for plant IT and quality teams.



- Raw/staging data is retained for replayability and auditability.
- Continuous telemetry is separated from discrete events and quality decisions.
- Qlik/Power BI can show the pilot dashboard while .NET/Python remains the intelligence engine.
- The customer can start with one route, one defect family and historical data.

Pilot packages customers can buy

Start with a low-friction assessment, then expand into correlation and prediction.

Data Readiness Assessment

1-2 weeks

Source list, data gaps, genealogy feasibility, pilot scope

Quality Correlation Diagnostic

3-6 weeks

Dashboard + suspected contributors for one route and one quality topic

Quality Risk Prediction Pilot

6-10 weeks

Baseline model, risk view, explanation and validation report

Production Deployment

3-6 months

Approved connectors, security, reports, governance and support

Most practical entry point: sell a Data Readiness Assessment, then convert to a Correlation Diagnostic Pilot.

What the customer sees

- Quality overview and defect trends
- Material genealogy and process history
- High-risk materials before the next step
- Top suspected contributors and evidence
- Data-quality warnings that explain why analytics may fail
- Recommended investigation actions for process experts

**The dashboard is the window.
The intelligence engine stays independent.**

Different from a normal BI dashboard

It is built for diagnosis, learning and prevention - not only reporting.

Normal dashboard

Shows KPI trends after the fact

Depends on manual interpretation

Works per report or system

Shows defect counts

Reporting focus

PlantProcess IQ

Connects process history with quality outcomes

Ranks suspected contributors and explains high-risk cases

Builds cross-system material/batch genealogy

Finds process windows and operating patterns linked to defects

Diagnosis, prediction and prevention focus

Start with one route. One defect. One clear business question.

PlantProcess IQ can begin with exported historical data, a short mapping workshop and a focused diagnostic pilot - without touching production control systems.

Data readiness

Mapping workshop

Correlation diagnostic

Risk prototype

Go / no-go roadmap

Offer: Data Readiness Assessment

PlantProcess IQ